

Wi-Fi Training Program

Module 4 - Assignment Answers

1. What is the Significance of the MAC Layer and in which Position is it in the OSI Model?

The Medium Access Control (MAC) layer is one of the two sublayers that make up the Data Link Layer (Layer 2) of the OSI (Open Systems Interconnection) model. It sits just above the Physical layer (Layer 1) and below the Logical Link Control (LLC) sublayer. The MAC layer acts as a critical bridge between the hardware-level transmission and higher-layer network protocols.

In terms of significance, the MAC layer governs how devices in a shared network medium gain access to that medium and transmit data. It ensures that multiple devices can coexist on the same wireless channel without consistently colliding or interfering with one another. The MAC layer is responsible for framing data received from higher layers into packets suitable for transmission over the physical medium.

In IEEE 802.11 wireless networks specifically, the MAC layer handles several critical operations including channel access coordination using the Distributed Coordination Function (DCF) and Point Coordination Function (PCF), acknowledgment of received frames, fragmentation and reassembly of data, encryption and security through protocols like WPA2/WPA3, power management for client devices, and association and authentication with access points.

The MAC layer is significant because without it, multiple wireless stations would transmit simultaneously, causing data corruption and channel chaos. It introduces order, reliability, and efficiency into what would otherwise be an uncoordinated shared medium. Its position at Layer 2 of the OSI model places it as the gatekeeper between raw bit transmission (Layer 1) and logical data delivery (Layer 3 and above).

2. Describe the Frame Format of the 802.11 MAC Header and the Purpose of Each Field

The IEEE 802.11 MAC frame consists of a header, a variable-length frame body, and a Frame Check Sequence (FCS). The MAC header contains the following fields:

Frame Control (2 bytes): This is the first and most critical field in the MAC header. It contains subfields that define the nature and behavior of the frame. It includes the Protocol Version (2 bits, always 00 for current standards), Type (2 bits, identifying Management, Control, or Data frames), Subtype (4 bits, further specifying frame types like Beacon, Probe, RTS, CTS, etc.), To DS and From DS flags (1 bit each, indicating whether the frame is headed to or from the distribution system), More Fragments flag (indicating if more fragments follow), Retry flag (indicating a retransmission), Power Management flag (indicating the sender's power state after the exchange), More Data flag (indicating buffered frames at the AP for a sleeping client), Protected Frame flag (indicating the frame body is encrypted), and the Order flag.

Duration/ID (2 bytes): This field is used for two purposes. In most frames, it carries the duration value in microseconds, which represents how long the channel will be occupied. Other stations use this value to update their Network Allocation Vector (NAV), a virtual carrier sensing mechanism. In Power Save Poll (PS-Poll) frames, this field carries the Association ID (AID) of the station.

Address 1 (6 bytes): This is the Receiver Address (RA). It represents the immediate next-hop recipient of the frame. In infrastructure mode, this is typically the MAC address of the access point or client that should immediately receive the frame.

Address 2 (6 bytes): This is the Transmitter Address (TA). It is the MAC address of the device that is transmitting the frame onto the wireless medium. This helps receiving devices know who sent the frame.

Address 3 (6 bytes): This field carries the Destination Address (DA) in frames going to the DS, or the Source Address (SA) in frames coming from the DS. It provides the final source or destination endpoint of the communication, which may differ from Address 1 and Address 2 in multi-hop or bridged scenarios.

Sequence Control (2 bytes): This field contains the Fragment Number (4 bits) and Sequence Number (12 bits). The sequence number uniquely identifies each MSDU (MAC Service Data Unit) and helps the receiver detect duplicate frames and reassemble fragments in the correct order.

Address 4 (6 bytes, optional): This field is present only in frames transmitted within a wireless distribution system (WDS) where both To DS and From DS flags are set. It carries the original source address when the frame is being forwarded between access points.

QoS Control (2 bytes, optional): Present in QoS data frames (802.11e and later), this field carries information about the traffic category, acknowledgment policy, TXOP (Transmission Opportunity) limit, and other quality-of-service parameters.

HT Control (4 bytes, optional): Introduced in 802.11n, this field carries high-throughput control information used for link adaptation, antenna selection, and other HT-specific functions.

Frame Check Sequence (4 bytes): Though technically at the end of the full frame rather than the header, the FCS is a CRC-32 checksum calculated over the entire MAC frame. The receiver recalculates it and compares with the transmitted value to detect any transmission errors.

3. List all the MAC Layer Functionalities Across Management, Control, and Data Planes

Management Plane

The management plane is responsible for establishing and maintaining wireless connections. Its key functionalities include:

Scanning: Stations perform active or passive scanning to discover available networks. Passive scanning involves listening for Beacon frames, while active scanning involves sending Probe Request frames and waiting for Probe Responses.

Authentication: Before associating, a station must authenticate with an access point. This involves either Open System Authentication (a simple two-frame

exchange) or Shared Key Authentication (a four-frame challenge-response mechanism using WEP, now deprecated).

Association and Reassociation: After authentication, a station sends an Association Request to the AP, which responds with an Association Response containing the AID. Reassociation is used when roaming between APs.

Disassociation and Deauthentication: These allow either the station or AP to gracefully terminate the connection.

Beaconing: APs periodically transmit Beacon frames to advertise their presence, SSID, supported rates, and other capabilities.

Probe Request and Response: Used during active scanning for AP discovery.

Action Frames: Used for a variety of management actions including spectrum management, QoS setup, block acknowledgment setup and teardown, radio measurement, and fast BSS transition.

Power Management Signaling: Management frames are used to signal the sleeping/active state of clients and to buffer frames at the AP.

Control Plane

The control plane manages access to the wireless medium and ensures reliable delivery of frames. Its functionalities include:

RTS/CTS (Request to Send / Clear to Send): These control frames implement a virtual carrier sense mechanism to prevent hidden node collisions. A station sends RTS, the AP responds with CTS, and other stations update their NAV accordingly.

ACK (Acknowledgment): After successfully receiving a unicast data or management frame, the receiver sends an ACK frame to confirm delivery. If no ACK is received, the sender retransmits.

Block ACK Request (BAR) and Block ACK (BA): Used in the block acknowledgment mechanism to confirm receipt of multiple aggregated frames at once.

PS-Poll: Used by sleeping clients to request buffered frames from the AP.

CF-End and CF-End+CF-Ack: Used to signal the end of a contention-free period in PCF operation.

NDP (Null Data Packet): Used in 802.11n/ac/ax for channel sounding and beamforming feedback.

Data Plane

The data plane carries user data and includes:

Data Frame Delivery: Basic transmission of MSDUs between stations and the AP.

Fragmentation and Reassembly: Large MSDUs are split into smaller fragments to improve reliability in noisy environments. Each fragment carries the same sequence number but an incrementing fragment number.

Encryption and Decryption: The MAC layer handles frame-level security using CCMP (AES-based, WPA2), TKIP (WPA), or GCMP (WPA3), including adding/removing the security header and MIC.

QoS Data Transmission: Using the Enhanced Distributed Channel Access (EDCA) mechanism with four access categories (Voice, Video, Best Effort, Background) for prioritized delivery.

Frame Aggregation: A-MSDU and A-MPDU aggregation techniques combine multiple frames into a single transmission to improve throughput.

MSDU Handling: The MAC layer receives MSDUs from the LLC sublayer, adds MAC headers and trailers, and prepares them for the physical layer.

4. Explain the Scanning Process and Its Types in detail

Scanning is the process by which a wireless station discovers available access points and the networks (BSSs) they provide. Scanning is performed when a station first powers on, when it roams, or when it needs to refresh its knowledge of available networks. The 802.11 standard defines two types of scanning:

Passive Scanning

In passive scanning, the station does not transmit any frames. Instead, it tunes to each channel and listens for Beacon frames periodically broadcast by access points. Beacons are transmitted at a fixed interval called the Target Beacon Transmission Time (TBTT), typically every 100 milliseconds (102.4 ms precisely).

The station must wait on each channel long enough to receive at least one Beacon. Since Beacons carry all the necessary information (SSID, BSSID, supported data rates, security capabilities, HT/VHT/HE capabilities, channel information, and timing synchronization), passive scanning allows a complete picture of available networks without the station revealing its presence.

Passive scanning is the preferred method for regulatory compliance in some countries where active transmission on certain channels (such as DFS channels) is restricted. However, passive scanning is slower because the station must wait the full Beacon interval on each channel.

Active Scanning

In active scanning, the station takes a more aggressive approach. It transmits Probe Request frames on each channel and waits for Probe Response frames from APs. A Probe Request may contain a specific SSID (directed probe) or a broadcast SSID (wildcard probe, which solicits responses from all APs on that channel).

When an AP receives a Probe Request, it responds immediately with a Probe Response frame containing essentially the same information as a Beacon. Active scanning is faster than passive scanning because the station does not need to wait for the next Beacon; it solicits an immediate response.

The station sets a MinChannelTime and MaxChannelTime for each channel. If no response is received by MinChannelTime, the station moves to the next

channel. If activity is detected before MinChannelTime expires, the station waits until MaxChannelTime before processing responses and moving on.

Active scanning is the default mode in most commercial devices due to its speed. However, it involves transmitting frames and is subject to regulatory restrictions on DFS channels. Some implementations use a hybrid approach, performing passive scanning on restricted channels and active scanning on others.

Channel Scanning Procedure

Regardless of the type, scanning involves the following steps: the station moves to a channel, performs the scan (passive or active), records found networks with their parameters (BSSID, SSID, RSSI, capabilities), moves to the next channel, and repeats until all required channels are scanned. The compiled list of APs is then used for the association process.

5. Brief about the Client Association Process

The client association process is the sequence of steps by which a wireless station establishes a connection with an access point and becomes a member of a BSS. The process involves the following Steps:

- Steps 1 - Scanning: The client first discovers available APs through passive or active scanning as described above. It compiles a list of candidate APs along with their BSSID, SSID, RSSI, and capability information.
- Steps 2 - AP Selection: Based on RSSI, security requirements, supported rates, and other factors, the client selects the most suitable AP from the discovered list.
- Steps 3 - Authentication (802.11 Open System): The client sends an Authentication Request frame to the selected AP. The AP responds with an Authentication Response frame indicating success or failure. In modern WPA2/WPA3 networks, this 802.11 authentication is purely an open system formality; actual security is handled at the higher layer (EAPOL 4-way handshake). Shared Key Authentication (WEP-based) is deprecated.

- Steps 4 - Association Request: After successful 802.11 authentication, the client sends an Association Request frame to the AP. This frame contains the client's supported capabilities including supported data rates, HT/VHT/HE capabilities, security requirements (RSN Information Element), listen interval (how often the client wakes to check for buffered data), and the SSID it wants to join.
- Steps 5 - Association Response: The AP evaluates the Association Request. If it accepts the client, it sends an Association Response containing a Status Code of 0 (success), an Association ID (AID) unique to this client within the BSS, and the AP's own supported capabilities.
- Steps 6 - 802.1X / EAPOL Authentication (for WPA2/WPA3 Enterprise): In enterprise networks, an EAP authentication exchange takes place between the client and the authentication server (via the AP), typically using RADIUS. The client proves its identity using credentials or certificates.
- Steps 7 - EAPOL 4-Way Handshake: Once the 802.11 association is complete and pre-shared key (PSK) or 802.1X credentials have been verified, the 4-way handshake is performed to derive and install encryption keys.
- Steps 8 - Data Exchange: After successful key installation, the client and AP can exchange encrypted data frames.

In roaming scenarios, the Reassociation Request and Response frames are used instead of Association Request/Response. These frames include the BSSID of the previous AP, allowing the new AP to potentially retrieve the client's state and enable faster transitions.

6. Explain each steps Involved in EAPOL 4-Way Handshake and the purpose of each Keys Derived from the process.

The EAPOL (Extensible Authentication Protocol over LAN) 4-Way Handshake is the mechanism used in WPA2 and WPA3 networks to mutually verify knowledge of the Pairwise Master Key (PMK) and to derive and install the session encryption keys. It occurs after 802.11 association and any prior 802.1X authentication.

Key Hierarchy Before the Handshake

The Pairwise Master Key (PMK) is derived either from the Pre-Shared Key (PSK) in personal mode (using PBKDF2 with SSID and passphrase) or from the EAP authentication in enterprise mode. The PMK is 256 bits long and is known to both the AP (Authenticator) and the client (Supplicant) before the handshake begins.

Message 1 (AP to Client)

The AP sends the first EAPOL-Key message containing ANonce (Authenticator Nonce), a 256-bit random value freshly generated by the AP for this session. At this point, the AP has not yet verified the client's PMK knowledge.

Upon receiving Message 1, the client generates its own SNonce (Supplicant Nonce) and uses both nonces along with the PMK, the AP's MAC address (Authenticator Address), and the client's MAC address (Supplicant Address) to derive the Pairwise Transient Key (PTK) using the PRF-X (Pseudo-Random Function) algorithm.

PTK Derivation and Components

$$\text{PTK} = \text{PRF-X}(\text{PMK}, \text{'Pairwise key expansion'}, \text{min}(\text{AA}, \text{SA}) \parallel \text{max}(\text{AA}, \text{SA}) \parallel \text{min}(\text{ANonce}, \text{SNonce}) \parallel \text{max}(\text{ANonce}, \text{SNonce}))$$

The PTK is divided into the following sub-keys: KCK (Key Confirmation Key, 128 bits) used to protect the integrity of EAPOL messages; KEK (Key Encryption Key, 128 bits) used to encrypt the GTK during the group key handshake; TK (Temporal Key, 128 bits or 256 bits) used as the actual data encryption key for unicast traffic between the client and AP; and in 802.11w (Management Frame Protection), the IGTK encryption and integrity keys are also derived.

Message 2 (Client to AP)

The client sends Message 2 containing the SNonce, the RSN Information Element (RSNIE) from the client's Association Request, and a MIC (Message Integrity Code) calculated over the entire EAPOL-Key frame using the KCK. The MIC proves to the AP that the client possesses the correct PMK (since only a device with the correct PMK can derive the correct KCK and compute a valid MIC).

The AP receives Message 2, extracts the SNonce, derives the PTK using the same inputs, and verifies the MIC using its derived KCK. If the MIC matches, the AP has confirmed that the client knows the PMK.

Message 3 (AP to Client)

The AP sends Message 3 containing the ANonce again (for integrity), the encrypted Group Temporal Key (GTK) wrapped using the KEK (via AES Key Wrap), the RSNIE, a sequence number to prevent replay attacks, and a MIC over the entire message using the KCK. The GTK is used to decrypt broadcast and multicast frames from the AP. In 802.11w networks, the encrypted IGTK is also included.

Message 4 (Client to AP)

The client installs the PTK and GTK, then sends Message 4 as an acknowledgment. Message 4 is a simple EAPOL-Key frame with a MIC, confirming to the AP that the client has received and installed the keys. Upon receiving Message 4, the AP installs the PTK on its side and marks the client as fully authenticated and connected.

Group Key Handshake

Separately from the 4-way handshake, a 2-message Group Key Handshake is performed whenever the GTK needs to be refreshed (e.g., when a client leaves the network). The AP sends the new encrypted GTK to all connected clients, and each client responds with an ACK.

Purpose Summary of Keys

PMK: The root key; not used directly for encryption. PTK: The per-session pairwise key hierarchy. KCK: Authenticates EAPOL messages. KEK: Encrypts GTK delivery. TK: Encrypts all unicast data frames. GTK: Decrypts broadcast and multicast frames from the AP.

7. Describe the power saving scheme in MAC layer and explore on the types of Power saving mechanisms

Power saving is a fundamental capability of the 802.11 MAC layer, allowing battery-powered devices to conserve energy by spending time in a low-power sleep state while still maintaining network connectivity. The MAC layer provides several power saving mechanisms:

Legacy Power Save Mode (PS Mode)

In legacy power save mode, a client notifies the AP of its intent to sleep by setting the Power Management bit in the Frame Control field. The AP then buffers any frames destined for the sleeping client. The AP periodically includes a Traffic Indication Map (TIM) information element in its Beacon frames. The TIM is a bitmap where each bit corresponds to an AID; a set bit indicates that the AP has buffered frames for that client.

The sleeping client wakes up before each Beacon, reads the TIM, and if its AID bit is set, sends a PS-Poll control frame to the AP. The AP responds with the buffered frame. The client may return to sleep immediately or stay awake to receive more frames (indicated by the More Data bit in the AP's response). If a DTIM (Delivery Traffic Indication Map) Beacon is approaching, multicast/broadcast frames are sent after the DTIM Beacon.

Legacy PS mode works but is inefficient due to the one-frame-per-PS-Poll exchange and the need to wake every Beacon interval.

Unscheduled Automatic Power Save Delivery (U-APSD)

U-APSD, introduced in 802.11e, improves upon legacy power save for QoS-enabled networks. Instead of the client polling for each frame, U-APSD defines trigger frames. When a client in power save mode sends a QoS data frame in a trigger-enabled access category, the AP takes this as a trigger to deliver all buffered downlink frames in that access category in a burst. The AP sets the More Data bit to indicate additional frames and sends an EOSP (End of Service Period) indicator when the burst is complete.

U-APSD significantly reduces the overhead of power save polling and is particularly well-suited for voice traffic, which has periodic, predictable uplink transmission.

Scheduled APSD (S-APSD)

S-APSD, also part of 802.11e, allows a fixed schedule for service periods to be negotiated between the client and AP. This is useful for predictable real-time traffic patterns such as voice calls, where both uplink and downlink occur at known intervals.

WMM Power Save (WMM-PS)

WMM-PS is the Wi-Fi Alliance certification that covers U-APSD functionality. It has become the standard power save mechanism for voice-capable Wi-Fi devices and is widely deployed in smartphones and VoIP endpoints.

Target Wake Time (TWT) - 802.11ax / Wi-Fi 6

TWT is the most advanced power saving mechanism, introduced in 802.11ax (Wi-Fi 6). TWT allows an AP to negotiate specific wake times (TWT sessions) with individual stations. Outside of these negotiated wake windows, the station can remain in deep sleep without needing to wake for every Beacon.

Individual TWT allows a station to negotiate a personalized schedule with the AP. Broadcast TWT allows the AP to group multiple clients with similar traffic patterns and wake them simultaneously, reducing overall channel contention.

TWT dramatically reduces power consumption for IoT devices and mobile clients, enabling sleep periods lasting seconds or even minutes. It is particularly impactful in dense deployments where many devices can sleep simultaneously.

Power Save Multi-Poll (PSMP)

PSMP, introduced in 802.11n, allows an AP to schedule both uplink and downlink transmission periods for multiple clients simultaneously. The AP announces a PSMP schedule, and clients wake only during their assigned slots. PSMP is rarely deployed in practice.

8. Describe the Medium Access Control Methodologies

The 802.11 MAC layer defines multiple methodologies for controlling access to the shared wireless medium. These are designed to prevent collisions and ensure fair and efficient use of the channel.

Distributed Coordination Function (DCF)

DCF is the fundamental mandatory access method for 802.11 networks and is based on Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA). Before transmitting, a station listens to the medium (physical and virtual carrier sense) and waits until the medium is idle for a period called the DIFS (DCF Interframe Space, 34 microseconds for 802.11g/n). After the DIFS, the station

enters a random backoff period, selecting a random number in the range $[0, CW-1]$ where CW is the Contention Window.

The backoff counter decrements by one for each idle slot time. The station transmits only when the counter reaches zero. If a collision is detected (no ACK received), the CW doubles up to a maximum (CW_{max}), and the process repeats. This binary exponential backoff prevents repeated collisions. After successful transmission, CW is reset to CW_{min} .

DCF also supports the optional RTS/CTS mechanism. A station sends an RTS frame, and upon receiving a CTS response, all other stations in range update their NAV (Network Allocation Vector) and refrain from transmitting for the duration specified. This addresses the hidden node problem.

Point Coordination Function (PCF)

PCF provides contention-free access for time-sensitive traffic. A Point Coordinator (PC), typically the AP, polls stations in sequence during a Contention-Free Period (CFP). The AP uses PIFS (PCF Interframe Space) to gain priority channel access. During the CFP, stations transmit only when polled. PCF alternates with DCF in a superframe structure. PCF is rarely implemented in practice and is effectively replaced by HCF.

Hybrid Coordination Function (HCF)

HCF, introduced in 802.11e, combines contention-based and contention-free access to provide Quality of Service. It consists of two access methods:

Enhanced Distributed Channel Access (EDCA): EDCA extends DCF by defining four Access Categories (ACs) with different priorities: AC_VO (Voice), AC_VI (Video), AC_BE (Best Effort), and AC_BK (Background). Each AC has its own AIFS (Arbitration Interframe Space), CW_{min} , CW_{max} , and TXOP (Transmission Opportunity) limit. Higher-priority ACs use shorter AIFS and smaller CW , giving them statistical priority over lower-priority traffic. EDCA enables differentiated QoS without a central coordinator.

HCF Controlled Channel Access (HCCA): HCCA provides polled, scheduled access managed by the Hybrid Coordinator (HC), typically the AP. The HC can grant TXOPs to stations based on negotiated traffic specifications (TSPEC). HCCA enables precise control over transmission timing for real-time applications. Like PCF, HCCA is rarely deployed in practice.

OFDMA-Based Channel Access (802.11ax)

802.11ax (Wi-Fi 6) introduces Orthogonal Frequency Division Multiple Access (OFDMA), which allows the AP to divide the channel into Resource Units (RUs) of varying sizes (26, 52, 106, 242, 484, 996 tones) and simultaneously schedule multiple clients in different RUs within a single TXOP. This dramatically improves spectral efficiency in dense environments. The AP manages OFDMA scheduling through Trigger frames, which inform clients of their assigned RUs and TXOP parameters. OFDMA can be used for both downlink (DL-OFDMA) and uplink (UL-OFDMA) transmissions.

9. Brief about the Block ACK Mechanism and Its Advantages

The Block ACK (Block Acknowledgment) mechanism was introduced in 802.11e and significantly improved throughput efficiency by reducing the overhead associated with per-frame acknowledgments. In the traditional immediate ACK scheme, every successfully received frame must be followed by an individual ACK, which is transmitted after a SIFS interval. For small frames or dense traffic, the ACK overhead becomes a substantial fraction of total airtime.

Block ACK Setup

Before using Block ACK, a Block ACK session must be established between two stations using an ADDBA (Add Block ACK) Request and Response exchange via Action frames. The ADDBA Request specifies the TID (Traffic Identifier), buffer size (number of frames the receiver can buffer), and Block ACK policy (immediate or delayed). The ADDBA Response confirms the session parameters.

Block ACK Operation

Once a Block ACK session is established, the transmitter sends a burst of frames (the burst size is limited by the TXOP or agreed buffer size) without waiting for individual ACKs between frames. After sending the burst, the transmitter sends a Block ACK Request (BAR) frame specifying the starting sequence number of the burst. The receiver responds with a Block ACK (BA) frame containing a bitmap (128 bytes) where each bit represents the reception status of a specific frame. A set bit indicates successful reception; a cleared bit indicates a missed frame, which the transmitter can then selectively retransmit.

Implicit vs Explicit Block ACK

In the explicit Block ACK scheme, the transmitter sends a BAR to trigger the BA. In the implicit (compressed) Block ACK scheme used in 802.11n and later, a compressed bitmap of 64 bits is used, reducing overhead. Some implementations also support the delayed Block ACK policy, where the receiver sends a regular ACK immediately and delivers the Block ACK later.

Advantages of Block ACK

Block ACK provides several important benefits. It reduces ACK overhead: instead of one ACK per frame, one BA covers an entire burst of frames, dramatically reducing the number of control frames on the air. It improves throughput: by eliminating the SIFS-ACK gap between every frame pair, more time is available for actual data transmission. It enables efficient aggregation: Block ACK is essential for A-MPDU aggregation to work effectively, since a single BA acknowledges all MPDUs in an aggregated frame. It supports selective retransmission: the bitmap in the BA allows only missed frames to be retransmitted rather than the entire burst, improving efficiency in lossy environments. It reduces latency: fewer round-trips are needed to deliver and confirm a batch of frames.

Block ACK Session Teardown

A Block ACK session is torn down using a DELBA (Delete Block ACK) Action frame, which can be sent by either party.

10. Explain about A-MSDU, A-MPDU, and A-MSDU in A-MPDU

Frame aggregation is one of the most impactful throughput enhancement mechanisms introduced in 802.11n and refined in subsequent standards. By combining multiple frames into a single large transmission, aggregation amortizes the overhead of headers, IFS periods, and acknowledgments over more payload data. Three aggregation formats are defined:

A-MSDU (Aggregate MAC Service Data Unit)

A-MSDU aggregates multiple MSDUs (upper layer data frames from the same source and destination MAC address pair and the same TID) into a single MPDU. Each MSDU is encapsulated in a subframe with a subframe header containing

destination address, source address, length, and optional padding to align the next subframe on a 4-byte boundary.

The aggregated A-MSDU is treated as a single MPDU by the MAC layer, meaning it has one MAC header, one FCS, and requires one ACK. The maximum A-MSDU size is 3,839 bytes (basic) or 7,935 bytes (large A-MSDU in 802.11n), and 11,454 bytes in 802.11ac/ax.

A-MSDU is efficient in low-error environments because if the A-MSDU is corrupted, the entire aggregated unit must be retransmitted since there is only one FCS protecting all subframes. It also requires all MSDUs to share the same receiver address, limiting its applicability in AP-to-multicast scenarios.

A-MPDU (Aggregate MAC Protocol Data Unit)

A-MPDU aggregates multiple MPDUs (each with its own MAC header and FCS) into a single PHY layer transmission (PPDU). Each MPDU is preceded by an MPDU delimiter (4 bytes) that contains the length of the MPDU, a CRC of the delimiter itself, and a reserved field.

Unlike A-MSDU, each constituent MPDU in an A-MPDU is independently delineated and checksummed. If one MPDU within the A-MPDU is corrupted, only that MPDU needs to be retransmitted; the others are accepted normally. This selective retransmission is made possible by the Block ACK mechanism, which reports the per-MPDU reception status in a bitmap.

A-MPDU is significantly more robust than A-MSDU in the presence of channel errors. The maximum A-MPDU size is 65,535 bytes in 802.11n, 1,048,575 bytes in 802.11ac (using extended A-MPDU), and 4,194,303 bytes in 802.11ax. A-MPDU aggregation is the primary aggregation method used in practice today.

A-MSDU in A-MPDU (Two-Level Aggregation)

802.11n and later standards support a two-level aggregation scheme where A-MSDUs are embedded within the MPDUs of an A-MPDU. In this scheme, each MPDU that forms part of the A-MPDU is itself an A-MSDU (i.e., it carries multiple MSDUs aggregated together).

This two-level approach combines the benefits of both methods. A-MSDU aggregation reduces MAC header overhead per MSDU, while A-MPDU

aggregation provides large transmission units with per-MPDU error recovery. The result is greater spectral efficiency because more user data fits into each PPDU while still maintaining the fault tolerance of MPDU-level FCS protection.

The tradeoff is increased complexity in the aggregation engine and potential retransmission of an entire A-MSDU (multiple MSDUs) if the containing MPDU is lost, rather than retransmitting individual MSDUs. Nevertheless, in high-throughput, low-error enterprise or home Wi-Fi environments, A-MSDU in A-MPDU delivers the best possible aggregation efficiency.

Comparison Summary

A-MSDU: Aggregates MSDUs into one MPDU; one FCS; all-or-nothing retransmission; low overhead per MSDU; limited by receiver address matching.
A-MPDU: Aggregates MPDUs into one PPDU; individual FCS per MPDU; selective retransmission via Block ACK; highly robust; most widely used in practice.
A-MSDU in A-MPDU: Two-level nesting; combines header efficiency of A-MSDU with robustness of A-MPDU; highest theoretical throughput in clean channels.